MATLIS DUALITY FORMULA AT A CLOSED POINT

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Allowed standard prerequisites (editorial).

- Dual of a perfect complex. more-algebra-lemma-dual-perfect-complex (cross-chapter, named). Moves the Hom into a tensor product.
- Dualizing complex. dualizing-lemma-dualizing-finite-length, dualizing-lemma-regular-gorenstein (cross-chapter, named). Identifies RHom(N, A) with the Matlis dual shifted.

Source reference. Original reference 0A7Q: focused-seeds/geometry/S9-eqv-duality-at-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AWX: focused-seeds/geometry/S9-eqv-duality-at-point/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07VI: focused-seeds/geometry/S9-eqv-duality-at-point/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be a scheme of finite type over k which is regular. Let $x \in X$ be a closed point. For a coherent $\mathcal{O}_X$-module $\mathcal{F}$ supported at x choose a coherent $\mathcal{O}_X$-module $\mathcal{F}'$ supported at x such that $\mathcal{F}_x$ and $\mathcal{F}'_x$ are Matlis dual. Then there is an isomorphism*

$$\mathrm{Hom}_X(\mathcal{F}, M) = H^0(X, M \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}'[-d_x])$$

where $d_x = \dim(\mathcal{O}_{X,x})$ functorial in M in $D_{perf}(\mathcal{O}_X)$.

Proof. Since $\mathcal{F}$ is supported at x we have

$$\mathrm{Hom}_X(\mathcal{F}, M) = \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, M_x)$$

and similarly we have

$$H^0(X, M \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}'[-d_x]) = \mathrm{Tor}_{d_x}^{\mathcal{O}_{X,x}}(M_x, \mathcal{F}'_x)$$

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Thus it suffices to show that given a Noetherian regular local ring A of dimension d and a finite length A-module N, if N' is the Matlis dual to N, then there exists a functorial isomorphism

$$\mathrm{Hom}_A(N, K) = \mathrm{Tor}_d^A(K, N')$$

for K in $D_{perf}(A)$. We can write the left hand side as $H^0(R\,\mathrm{Hom}_A(N, A) \otimes^{\mathbf{L}}_A K)$ by More on Algebra, Lemma 07VI and the fact that N determines a perfect object of $D(A)$. Hence the formula holds because

$$R\,\mathrm{Hom}_A(N, A) = R\,\mathrm{Hom}_A(N, A[d])[-d] = N'[-d]$$

by Dualizing Complexes, Lemma 0A7Q and the fact that $A[d]$ is a normalized dualizing complex over A (A is Gorenstein by Dualizing Complexes, Lemma 0AWX). $\square$